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Essential Histology

A Concept-Based Guide for PGME

AREMS-HY — Academic Series

Primary Reference: Junqueira's Basic Histology: Text and Atlas — 17th Edition, by Anthony L. Mescher, McGraw Hill

Official Syllabus: Junqueira 17th Chapters 1,2,3,4,5,6,9,11,12,13,14,15,16,17,18,20 — 16 chapters

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****Official Syllabus (Specialty Examination):**** Based on Junqueira 17th Chapters 1,2,3,4,5,6,9,11,12,13,14,15,16,17,18,20 — total 16 chapters. All study materials based primarily on this edition and remain within official syllabus.

Preface

This book teaches histology as a logical system, not a list of facts.

Every tissue has a reason for being built the way it is. If you understand the reason, you remember the structure. If you understand the structure, you can recognize it, distinguish it, and predict what happens when it fails.

The book is aligned with Junqueira's Basic Histology, 17th Edition as primary and authoritative reference, and organized for the Afghanistan 1405 Clinical Medicine postgraduate examination official syllabus (16 chapters). It is not a compressed copy of Junqueira. It is a learning system that makes required histology understandable, memorable, connected, and teachable.

Each chapter opens with a question, gives you a map, builds the concept from components to complete tissue, connects structure to function, teaches you how to recognize and distinguish, explains clinical meaning, and ends with what you must know and whether you can explain it without the book.

Read in order. Each chapter grows from the previous one.

How to Use This Book

****First pass — map:**** Read Opening Question, Why This Matters, Learning Objectives, Big Picture, Core Concept.

****Second pass — build:**** Work through Build the Concept, Structure, Function, Structure→Function, Classification.

****Third pass — discriminate:**** Study Compare & Distinguish and Recognition Logic. These answer "How do I know which one I am looking at?"

****Before exam — consolidate:**** Use High-Yield, Integrated Summary, Rapid Review. Test yourself with Mastery Check questions (open-ended, not MCQs).

Question Bank is separate: Essential Histology: Question Bank — questions traceable to learning objectives, based on official 16-chapter syllabus.

Contents — Official 16-Chapter Syllabus

1. Chapter 1 — Histology & Its Methods of Study (Junqueira Ch1)
2. Chapter 2 — The Cytoplasm (Junqueira Ch2)
3. Chapter 3 — The Nucleus (Junqueira Ch3)
4. Chapter 4 — Epithelial Tissue (Junqueira Ch4)
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7. Chapter 7 — Nerve Tissue & the Nervous System (Junqueira Ch9)
8. Chapter 8 — The Circulatory System (Junqueira Ch11)
9. Chapter 9 — Blood (Junqueira Ch12)
10. Chapter 10 — Hemopoiesis (Junqueira Ch13)
11. Chapter 11 — The Immune System & Lymphoid Organs (Junqueira Ch14)
12. Chapter 12 — Digestive Tract (Junqueira Ch15)
13. Chapter 13 — Organs Associated with the Digestive Tract (Junqueira Ch16)
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Chapter 1 — Histology & Its Methods of Study

Junqueira 17th Edition — Chapter 1

Opening Question

How does a pathologist turn a lump of tissue into a diagnosis you can see under a microscope — and why does the choice of stain decide what you will find?

Why This Matters

Every biopsy is a question: which molecule is accumulating, which cell is proliferating, which barrier is broken? No single technique shows everything. The fixative, the solvent, the dye, and the microscope each reveal one property and hide others. Understanding this logic prevents misdiagnosis and tells you which stain to order next.

A 45-year-old man with hepatomegaly could have lipid, glycogen, iron, or amyloid accumulation — four diseases, four different stains, four different treatments.

Learning Objectives

By the end of this chapter, you will be able to:

1. Explain the physical and chemical purpose of each step in tissue processing and predict artifact when a step is skipped or delayed.
2. Derive why nucleus stains blue and cytoplasm pink in H&E from dye charge chemistry.
3. Match special stains to molecular targets (PAS, silver, trichrome, Orcein, Oil Red O, Congo red, Prussian blue, Von Kossa, Alcian blue, Feulgen) and state processing requirements.
4. Select appropriate microscopy modality for a stated investigative goal (live cells, thick specimen, surface topography, nanometer detail, membrane faces, gene copy number).
5. Explain why light resolution is limited to $\sim 0.2 \mu\text{m}$ and why electron microscopy exceeds it.
6. Interpret staining patterns (basophilia, acidophilia, metachromasia, birefringence, empty vacuoles) as molecular content.
7. Distinguish frozen section from paraffin by what each preserves, sacrifices, and is used for.

Big Picture

A tissue is transparent and fragile. Histology solves two problems:

1. ****Make it hard enough to cut**** — fixation → dehydration → clearing → embedding
2. ****Create contrast**** — charge (H&E), oxidation (PAS, Feulgen), metal binding (silver), solubility (lipid dyes), light interaction (birefringence, fluorescence, electron scatter)

No method is universal. Each technique = one type of contrast. If a structure is absent, ask whether the method removed it.

Core Concept

****Each technique reveals primarily the specific property it was designed to detect; no method is universal.**** The result must be interpreted as chemistry, not as magic color.

****CORE IDEA:**** Contrast is not decoration. It is chemistry made visible — each stain and microscope reveals a specific chemical or physical property.

Build the Concept

****Living tissue autolyzes.**** Enzymes digest the tissue within minutes after blood supply stops. Fixation arrests this.

****Fixation:**** cross-links macromolecules.

- ****Formalin (formaldehyde):**** reacts with amino/imino groups → methylene bridges. Standard for light microscopy. Preserves proteins adequately, does not preserve lipids well, moderate ultrastructure.

- ****Glutaraldehyde:**** dialdehyde, faster and more extensive cross-linking. Standard primary fixative for electron microscopy.

- ****Osmium tetroxide:**** binds phospholipid head groups of membranes, fixing lipids, and as a heavy metal scatters electrons → provides contrast in TEM. It is both fixative and contrast agent.

****Dehydration:**** water removed by ascending ethanol series (70% → 95% → 100%). Required because paraffin is hydrophobic.

****Clearing:**** ethanol removed by xylene (or substitute). Xylene is miscible with both ethanol and paraffin, making tissue translucent (cleared) and paraffin-miscible.

****Embedding:**** paraffin infiltrates and hardens tissue into cuttable block.

****Sectioning:**** paraffin sections ~5 μm (range ~1-10 μm). Semi-thin resin sections for EM orientation ~1 μm stained with toluidine blue. Ultrathin EM sections ~50-90 nm.

****Staining:**** creates contrast.

Structure — The Staining Logic

****H&E — charge-based:****

- ****Hematoxylin:**** basic dye, positively charged. Binds negatively charged phosphate backbone of DNA/RNA → blue/purple. Term: ****basophilic.****
- ****Eosin:**** acidic dye, negatively charged. Binds positively charged basic side chains (lysine, arginine) of proteins and collagen → pink/red. Term: ****acidophilic/eosinophilic.****
- Deliberate inversion that confuses students: basophilic structures are acidic (they attract basic dye).

****PAS — oxidation:**** periodic acid oxidizes hexose glycols to aldehydes → Schiff reagent binds → magenta. PAS is primarily used to demonstrate carbohydrate-rich structures — glycogen, mucin, basement membrane glycoproteins (laminin, perlecan, entactin associated with type IV collagen network), fungal walls — and is not a collagen-specific stain. Basement membranes are PAS-positive because of their carbohydrate-rich components, not because PAS stains collagen directly.

****Silver impregnation — argyrophilia:**** reticular fibers (type III collagen) reduce silver salts → black. Reticular fiber = type III collagen in delicate stroma (liver, lymphoid organs, endocrine glands).

****Masson trichrome — differential uptake:**** collagen blue/green vs muscle/cytoplasm red. Fibrosis assessment.

****Orcein / Verhoeff — elastic:**** elastic fibers (elastin core + fibrillin microfibrils) → brown-black.

****Lipid dyes — solubility:**** Oil Red O, Sudan black. Lipid dissolves in ethanol/xylene, so routine paraffin processing removes it, leaving empty vacuoles. To demonstrate lipid, need frozen section (no solvents) + lipophilic dye → red/black droplets.

****Congo red — orientation:**** amyloid β -sheet fibrils bind Congo red and orient dye molecules → apple-green birefringence under polarized light. Birefringence alone is not diagnostic; combination of Congo red + green birefringence is.

****Prussian blue (Perls) — chemical reaction:**** ferric iron + potassium ferrocyanide → ferric ferrocyanide (blue precipitate). Detects hemosiderin.

****Von Kossa — indirect mineral demonstration:**** silver nitrate reacts with phosphate and carbonate anions that are bound to calcium in mineralized deposits. Silver substitutes, then reduced to metallic silver → black. It does NOT stain calcium ion directly; it demonstrates sites of mineralization where calcium phosphate/carbonate has deposited. Use as indirect indicator of calcification.

****Alcian blue — acidic mucin:**** sulfated and carboxylated glycosaminoglycans → blue.

****Feulgen — DNA:**** acid hydrolysis exposes aldehydes on deoxyribose → Schiff → magenta. Specific for DNA amount.

Function

Each stain answers one question: where is nucleic acid, protein, carbohydrate, reticular fiber, collagen vs muscle, elastic fiber, lipid, amyloid, iron, mineralization, acidic mucin, DNA?

Structure → Function

Why does blue cytoplasm mean protein synthesis? Plasma cell cytoplasm is basophilic because it is packed with rough ER and ribosomes — RNA phosphate negative → binds hematoxylin blue. Muscle fiber is eosinophilic because it is packed with contractile proteins — basic residues positive → binds eosin pink. Staining reflects dominant macromolecule.

Classification

****By mechanism:****

- Charge: H&E
- Oxidation→Schiff: PAS, Feulgen
- Metal reduction: silver, Von Kossa
- Lipophilic: Oil Red O
- Birefringence: Congo red + polarizer, collagen intrinsic
- Immunologic: IHC, immunofluorescence
- Nucleic acid hybridization: FISH
- Isotope incorporation: autoradiography

****Microscopy by contrast:****

- Phase-contrast/DIC: refractive index → live unstained cells
- Dark-field: scattered light → bright object on black background (spirochetes)
- Fluorescence: fluorochrome excitation/emission → specific molecules, DAPI DNA
- Confocal: pinhole rejects out-of-focus light → crisp optical section of thick specimen
- Polarizing: birefringence → amyloid, collagen, crystals
- TEM: electron scatter → internal ultrastructure 2-D, ~0.1 nm resolution
- SEM: secondary electrons → 3-D surface topography
- Freeze-fracture: fracture through hydrophobic interior of lipid bilayer → E-face (outer leaflet inner aspect) and P-face (inner leaflet outer aspect) with intramembrane particles

Compare & Distinguish

Recognition Logic

****Look for...**** blue nucleus + pink cytoplasm = H&E baseline.

****Then confirm with...**** magenta = PAS carbohydrate; black network = silver reticular; blue/green vs red = trichrome collagen vs muscle; red droplets on frozen = lipid; apple-green birefringence after Congo red under polarized light = amyloid; blue granules = iron Perls; black deposits after silver nitrate = mineralization (Von Kossa, indirect).

****Do not confuse with...**** metachromasia (toluidine blue → purple/red on dense polyanions like heparin in mast cells due to dye polymerization) is not basophilia. Birefringence of collagen (white) is not Congo red green.
****Decisive feature is...**** charge chemistry for H&E, oxidation for PAS, silver reduction for reticular and mineralization, solubility for lipid, polarization for amyloid.

Clinical Correlation

****Fatty liver vs glycogen storage vs hemochromatosis vs amyloidosis:**** same hepatomegaly, different stain. Fat needs frozen + Oil Red O because routine processing dissolves lipid → empty vacuoles on H&E. Glycogen needs PAS (magenta) and diastase sensitivity. Iron needs Prussian blue. Amyloid needs Congo red + polarized apple-green. Von Kossa would show black only if calcification present, and indicates phosphate/carbonate of mineral, not calcium ion itself.
****Frozen section:**** surgeon needs margin in 15 minutes. Paraffin cannot be done that fast. Frozen short-circuits dehydration/clearing/embedding, preserves enzymes/lipids, but ice crystals create artifacts.
****Goodpasture/Alport:**** basement membrane best shown by PAS (glycoproteins type IV collagen + laminin). Thickened in diabetes.

Common Misconceptions

****Misconception:**** "Clearing removes water."
****Reality:**** Dehydration (ethanol) removes water. Clearing (xylene) removes ethanol and makes tissue miscible with paraffin. Each step exists because no single reagent bridges water and paraffin.
****Misconception:**** "Von Kossa stains calcium."
****Reality:**** Von Kossa demonstrates phosphate and carbonate anions in mineralized matrix via silver substitution. Calcium is inferred because calcium phosphate/carbonate is the mineral, but the reaction detects anion, not calcium ion. Always report as "mineralization" or "calcium phosphate deposits, Von Kossa positive (indirect)".
****Misconception:**** "More magnification = more resolution."
****Reality:**** Resolution = minimum distance two points can be separated as distinct, set by wavelength and numerical aperture. Light limited to ~0.2 μm by visible light wavelength (~400-700 nm). Electron wavelength much shorter → ~0.1 nm. Magnification without resolution is empty magnification.
****Misconception:**** "H&E shows everything."
****Reality:**** H&E shows charge distribution only. Lipid, glycogen amount, reticular fibers, elastic fibers, iron, mineral, amyloid need special stains.

High-Yield Knowledge

****Must Know:****
- Basophilic = nucleic acid = binds basic dye hematoxylin = blue. Acidophilic = protein = binds acidic dye eosin = pink.
- PAS = carbohydrate (glycogen, mucin, basement membrane, fungi) — not collagen, not reticular.

- Reticular fibers = type III collagen = silver black.
- Collagen vs muscle = Masson trichrome.
- Elastic fibers = Orcein/Verhoeff.
- Lipid = frozen + Oil Red O (routine processing dissolves it).
- Amyloid = Congo red + apple-green birefringence polarized.
- Iron = Prussian blue. Mineralization = Von Kossa (phosphate/carbonate, silver black, indirect indicator of calcium).
- LM fixative = formalin. EM = glutaraldehyde + osmium (osmium = membranes + contrast).
- Light resolution $\sim 0.2 \mu\text{m}$, EM $\sim 0.1 \text{ nm}$ (shorter wavelength).
- **Must Distinguish:****
- Basophilic vs metachromatic (blue vs purple shift due to polymerization)
- Silver reticular (type III) vs trichrome collagen (type I) vs Orcein elastic
- TEM inside vs SEM surface vs freeze-fracture membrane faces
- Frozen (lipid, enzymes, speed, inferior morphology) vs paraffin (best morphology, lipid lost)
- **Must Understand:****
- Why empty vacuole on H&E does not mean empty cell — lipid dissolved.
- Why Von Kossa is indirect.
- Why resolution $\neq$ magnification.

Integrated Summary

Histology makes transparent fragile tissue cuttable and contrasted. Processing is miscibility ladder: water $\rightarrow$ alcohol $\rightarrow$ xylene $\rightarrow$ paraffin, each step bridging to next; fixation cross-links to prevent autolysis. Staining is chemistry: charge, oxidation, metal reduction, solubility, birefringence, antibody-antigen, probe hybridization, isotope incorporation. Microscopy is optics: refractive index, fluorescence, birefringence, electron scatter. Choose fixative, stain, microscope to answer specific molecular question. No technique universal.

Mastery Check

1. Explain why dehydration and clearing are two separate steps, not one.
2. A plasma cell is intensely basophilic while skeletal muscle is eosinophilic — explain from dominant macromolecule and dye chemistry.
3. A researcher wants to prove active DNA synthesis vs amount of DNA vs cycling marker — compare ^3H -thymidine autoradiography, Feulgen, Ki-67 IHC.
4. Why is TEM required to see podocyte foot process effacement in minimal change disease?
5. How do you distinguish amyloid birefringence from collagen birefringence?
6. Why does freeze-fracture split membrane into two faces rather than separating membrane from cytoplasm?
7. A signet-ring cell with single large clear space and flattened nucleus on H&E — most safe interpretation and how to confirm?

Transition

You have now seen how tissue is prepared and stained. Next, step inside the cell itself — the machinery that every stain in this chapter is lighting up: the cytoplasm.

Rapid Review

- Processing: fixation (formalin LM, glutaraldehyde+osmium EM) → dehydration (ethanol removes water) → clearing (xylene removes ethanol) → embedding (paraffin) → ~5 μm section.
- H&E: hematoxylin basic + binds DNA/RNA phosphate → blue basophilic; eosin acidic – binds protein basic residues → pink acidophilic.
- PAS magenta = carbohydrate. Silver black = reticular type III. Trichrome blue/green collagen vs red muscle. Orcein = elastic. Oil Red O red = lipid, needs frozen. Congo red + polarized green = amyloid. Perls blue = iron. Von Kossa black = mineral phosphate/carbonate indirect.
- Microscopes: phase live, confocal pinhole thick, TEM internal 0.1 nm, SEM surface, freeze-fracture E/P faces.
- Frozen = speed + lipid + enzymes, inferior morphology. Paraffin = best morphology, lipid lost.

Chapter 2 — The Cytoplasm

Junqueira 17th Edition — Chapter 2

Opening Question

How does a cell build, move, and ship thousands of proteins without mixing them up?

Why This Matters

Pancreatic acinar cells secrete amylase, plasma cells secrete antibody, liver cells detoxify drugs — same basic machinery, different outputs. Kartagener syndrome (immotile cilia), I-cell disease (lysosomal enzymes misrouted), and microvillus inclusion disease all trace to cytoplasmic machinery failure.

Learning Objectives

1. Describe membrane flow from RER → Golgi → vesicles and explain how signal sequences and M6P targeting determine destination.
2. Distinguish RER, SER, Golgi, lysosome, peroxisome, mitochondria by structure, function, and microscopic appearance.
3. Classify cytoskeletal filaments by diameter, protein, stability, and function (microfilaments actin 6-8 nm, intermediate 10 nm, microtubules 25 nm).
4. Explain why basophilic cytoplasm indicates RER-rich protein synthesis.
5. Predict consequence of defects in microtubule motors, dynein arms, lysosomal targeting, peroxisomal oxidases.

Big Picture

Cytoplasm = cytosol (aqueous phase) + organelles (membrane-bound compartments) + cytoskeleton (scaffolding and tracks) + inclusions (stored products).

Governing principle: ****compartmentalization by membrane + transport by vesicles + positioning by cytoskeleton.****

Core Concept

The cell solves crowding by membranes that create distinct chemical environments, and solves movement by cytoskeletal tracks and motors.

****CORE IDEA:**** The cytoplasm organizes work by compartment: RER synthesizes proteins for export, SER synthesizes lipids and handles detoxification and calcium, Golgi modifies and sorts, lysosomes digest, peroxisomes oxidize, mitochondria produce ATP, and the cytoskeleton organizes space and transport.

Build the Concept

****RER:**** studded with ribosomes (rRNA basophilic). Co-translational insertion: signal sequence → SRP → translocon → lumen. Protein synthesis for secretion, lysosomal enzymes, membrane proteins. Extensive in protein-secreting cells (pancreas, plasma cell) → basophilic cytoplasm.

****SER:**** no ribosomes. Lipid synthesis, steroid synthesis (adrenal cortex fasciculata, Leydig), detoxification (smooth ER proliferation with drugs, cytochrome P450), calcium storage (muscle sarcoplasmic reticulum).

****Golgi:**** cis (receiving, near RER) → medial → trans (shipping) + TGN (trans-Golgi network). Glycosylation, sulfation, sorting: M6P for lysosomal enzymes, signal peptides for secretion. Polarized: cis convex, trans concave with vesicles.

****Lysosome:**** acid hydrolases (acid phosphatase marker) active at ~pH 5, proton pump. Primary (enzyme only) + secondary (enzyme + substrate). M6P receptor targets enzymes to lysosome; defect → I-cell disease enzymes secreted extracellularly, inclusions.

****Peroxisome:**** oxidase + catalase. Beta-oxidation of very long chain fatty acids, detox H₂O₂, not acid hydrolases. Distinct from lysosome. Marker: catalase. Defect → Zellweger.

****Mitochondria:**** double membrane, outer smooth, inner cristae (tubular in steroid cells), matrix with enzymes, mtDNA, ribosomes. ATP synthesis, apoptosis, calcium. Eosinophilic cytoplasm when abundant.

****Cytoskeleton:****

- ****Microfilaments:**** ~6-8 nm, actin + myosin, dynamic, cortex, microvilli core, contractile ring, focal adhesions. Polarity barbed/pointed, ATP.

- ****Intermediate filaments:**** ~10 nm, stable, cell-type specific: keratin (epithelium), vimentin (mesenchyme), desmin (muscle), neurofilament (neuron), GFAP (astrocyte), lamins (nuclear). Anchors desmosomes and hemidesmosomes.

- ****Microtubules:**** ~25 nm, tubulin α/β dimer, dynamic instability, GTP, MTOC centrosome, 9+2 in cilia, tracks for kinesin (anterograde) and dynein (retrograde). Mitotic spindle.

Structure → Function

Why does pancreatic acinar cell have basal basophilia and apical eosinophilia? Basal RER (basophilic) synthesizes zymogen, apical zymogen granules (eosinophilic protein) store enzymes. Polarity reflects flow.

Why does microvillus have actin core while cilium has microtubule core? Microvillus needs stiff stable scaffold to increase surface area without movement → actin. Cilium needs motility → microtubule doublet sliding via dynein.

Classification

Organelles by membrane: single membrane (RER, SER, Golgi, lysosome, peroxisome) vs double (mitochondria, nucleus). Filaments by size and protein.

Compare & Distinguish

Recognition Logic

****Look for...**** basal blue cytoplasm + apical pink granules = protein-secreting cell (RER → zymogen). Central dark nucleus + basophilic cytoplasm + clock-face chromatin = plasma cell (RER-rich antibody).

****Then confirm with...**** microvilli (actin, no motility, brush border) vs cilia (9+2 microtubule, motile) vs stereocilia (long branched microvilli, not true cilia, epididymis). Terminal web (actin) under microvilli.

****Do not confuse with...**** RER basophilia (ribosomes) vs heterochromatin basophilia (DNA). Both blue but location and cell type differ.

****Decisive feature:**** ~6-8 nm actin vs ~10 nm intermediate vs ~25 nm tubulin; 9+2 microtubule for motile cilia; M6P for lysosome targeting.

Clinical Correlation

****Kartagener syndrome:**** dynein arm defect → immotile cilia → bronchiectasis, sinusitis, situs inversus (nodal cilia defect). Histology: cilia present but not motile, EM shows missing dynein arms.

****I-cell disease (mucopolidosis II):**** GlcNAc phosphotransferase defect → no M6P tag → lysosomal enzymes not targeted to lysosome → secreted, lysosomes lack enzymes → inclusion bodies (phase-dense inclusions).

****Microvillus inclusion disease:**** actin cytoskeleton defect → atrophic microvilli.

Common Misconceptions

****Misconception:**** "SER is only for detox."

****Reality:**** SER also synthesizes lipid, steroid, stores Ca^{2+} . In muscle, SER is sarcoplasmic reticulum Ca^{2+} release for contraction.

****Misconception:**** "Lysosome and peroxisome are similar because both digest."

****Reality:**** Lysosome digests via acid hydrolases, acidic pH, M6P targeted from Golgi. Peroxisome oxidizes via oxidase/catalase, neutral, ER-derived, handles very long chain

fatty acids and H₂O₂.

****Misconception:**** "All basophilia is nucleus."

****Reality:**** Cytoplasmic basophilia = RER/ribosomes = protein synthesis.

High-Yield

****Must Know:**** RER basophilic protein synthesis, SER lipid/detox/Ca²⁺, Golgi cis→trans glycosylation + sorting, lysosome acid hydrolase M6P, peroxisome oxidase/catalase, mitochondria double membrane cristae. Filament diameters ~6-8, ~10, ~25 nm. Actin microvilli, tubulin cilia 9+2.

****Must Distinguish:**** RER vs SER, lysosome vs peroxisome, microvilli actin vs cilia tubulin vs stereocilia long microvilli.

****Must Understand:**** Membrane flow polarity, why basophilic cytoplasm means synthetic activity, why cytoskeleton cell-type specific intermediate filaments are diagnostic in pathology.

Integrated Summary

Cytoplasm organizes chemistry by membranes: RER makes proteins for export/membrane/lysosome, SER makes lipid and handles calcium/detox, Golgi modifies and sorts via M6P and signal peptides, lysosome digests at acidic pH, peroxisome oxidizes via catalase, mitochondria makes ATP. Cytoskeleton provides scaffold and tracks: actin for cortex and microvilli, intermediate filaments for mechanical stability and cell identity, microtubules for transport and cilia. Structure predicts staining and function.

Mastery Check

1. Explain why plasma cell cytoplasm is intensely basophilic and where antibody goes after synthesis.
2. Reconstruct membrane flow from signal sequence to secreted protein, including M6P branch to lysosome.
3. Compare microvillus, cilium, stereocilium by core protein, motility, location.
4. Predict consequence of kinesin vs dynein motor defect.
5. Why does I-cell disease cause extracellular lysosomal enzymes and intracellular inclusions?

Transition

The cytoplasm builds and moves. The nucleus controls what is built, when, and how much. Next, how two meters of DNA fits in a six micrometer nucleus and remains readable.

Rapid Review

- RER basophilic = protein for export/membrane/lysosome. SER = lipid, steroid, detox, Ca²⁺.
- Golgi cis→trans, glycosylation, sorting, M6P lysosome.
- Lysosome acid hydrolase pH5, peroxisome oxidase/catalase.
- Filaments: actin ~6-8 nm microvilli, intermediate ~10 nm keratin/vimentin/desmin/neurofilament/GFAP, microtubule ~25 nm tubulin 9+2 cilia.

- Kartagener dynein immotile cilia, I-cell M6P defect.

Chapter 3 — The Nucleus

Junqueira 17th Edition — Chapter 3

Opening Question

How does 2 meters of DNA fit in a 6 μm nucleus and still remain readable?

Why This Matters

Cancer diagnosis rests on nuclear atypia: irregular envelope, coarse chromatin, prominent nucleoli. Barr body (inactive X) explains dosage compensation. Understanding euchromatin vs heterochromatin explains why some cells transcribe actively and others do not.

Learning Objectives

1. Describe nuclear envelope (double membrane, perinuclear space, pores ~ 120 MDa, lamina) and its functions.
2. Distinguish euchromatin (active, light, dispersed) vs heterochromatin (inactive, dark, condensed) functionally and microscopically.
3. Explain nucleolus organization (FCC, DFC, GC) and its role in rRNA transcription and ribosome biogenesis.
4. Predict consequence of lamin defects and nuclear pore transport defects.
5. Recognize nuclear features of protein-synthesizing vs inactive cells.

Big Picture

Nucleus = genome storage + reading control. Envelope = barrier + transport. Chromatin = DNA + histones, organization around accessibility. Nucleolus = ribosome factory.

Core Concept

****Chromatin organization controls access to genetic information.**** Euchromatin open for transcription, heterochromatin closed.

****CORE IDEA:**** A pale, euchromatic nucleus with a prominent nucleolus indicates active transcription and ribosome production, while a dark, heterochromatic nucleus indicates quiescence.

Build the Concept

****Nuclear envelope:**** double membrane, outer continuous with RER, perinuclear space, inner with lamina (lamin intermediate filaments) for mechanical support. Pores: large ~ 120 MDa complexes, selective transport, importin/exportin, Ran-GTP.

****Chromatin:**** DNA + histone octamer $\rightarrow$ nucleosome $\rightarrow$ 11 nm fiber $\rightarrow$ 30 nm fiber $\rightarrow$ looped domains. Euchromatin: dispersed, transcriptionally active, light in LM, peripheral

decondensed. Heterochromatin: condensed, inactive, dark, often peripheral under envelope and around nucleolus. Constitutive (always condensed, centromere) vs facultative (inactive X Barr body).

****Chromosome:**** condensed chromatin during mitosis, centromere, telomere.

****Nucleolus:**** not membrane-bound, organizer regions (NOR) contain rDNA. Zones: fibrillar center (rDNA), dense fibrillar component (transcription), granular component (processing). Prominent in protein-synthesizing cells (plasma cell, neuron). Multiple nucleoli in growing cells.

Structure → Function

Why does active cell have large pale nucleus with prominent nucleolus? Active transcription needs dispersed euchromatin (light) and ribosome biogenesis (nucleolus prominent). Inactive lymphocyte has small dark heterochromatic nucleus with no nucleolus — minimal transcription.

Why does nuclear envelope have pores? Nucleus needs to export mRNA and ribosomal subunits and import proteins and transcription factors, but keep DNA separate. Pores provide selective gated transport.

Classification

Chromatin: euchromatin vs heterochromatin (constitutive vs facultative). Nucleolus: 1-2 prominent in active cells, inconspicuous in inactive.

Compare & Distinguish

Recognition Logic

****Look for...**** pale nucleus + prominent nucleolus = active protein synthesis (plasma cell, neuron, hepatocyte). Dark small nucleus = inactive (lymphocyte, sperm).

****Then confirm with...**** Barr body at periphery in female somatic cells = inactive X, heterochromatin.

****Do not confuse with...**** nucleolus vs inclusion: nucleolus inside nucleus, inclusion in cytoplasm.

****Decisive feature:**** euchromatin light central, heterochromatin dark peripheral, nucleolus prominent when RER-rich cytoplasm basophilic — coupled.

Clinical Correlation

****Cancer atypia:**** irregular nuclear envelope, coarse clumped chromatin, multiple prominent nucleoli, high N:C ratio. Reflects abnormal transcription and proliferation.

****Laminopathies:**** lamin A/C mutations → muscular dystrophy, progeria — nuclear fragility.

****Barr body:**** buccal smear, sex chromatin, dosage compensation.

Common Misconceptions

****Misconception:**** "Heterochromatin is junk."

****Reality:**** Heterochromatin is inactive but functional: centromere stability, dosage compensation (Barr body), gene silencing.

****Misconception:**** "Nucleolus is membrane-bound organelle."

****Reality:**** Nucleolus is non-membrane compartment formed around NORs, dynamic.

High-Yield

****Must Know:**** Envelope double + pores 120 MDa + lamina lamin; euchromatin light active, heterochromatin dark inactive; nucleolus rRNA factory prominent in protein synthesis; Barr body inactive X peripheral.

****Must Distinguish:**** Euchromatin vs heterochromatin, nucleolus vs inclusion.

****Must Understand:**** Why active cells have pale nuclei with prominent nucleoli and basophilic cytoplasm — coupled transcription and translation demand.

Integrated Summary

Nucleus stores genome in chromatin whose condensation controls access: euchromatin open and light for active transcription, heterochromatin closed and dark for silencing. Envelope double membrane with selective pores and lamin support separates but connects to cytoplasm. Nucleolus around rDNA makes rRNA and ribosomal subunits, prominent when protein synthesis high. Nuclear morphology predicts cell activity.

Mastery Check

1. Explain why small lymphocyte nucleus is dark and plasma cell nucleus is pale with prominent nucleolus.
2. Describe nuclear pore function and why it needs to be large and selective.
3. What is Barr body and why at periphery?
4. How does lamin defect cause disease?

Transition

Feature	Frozen Section	Paraffin Section
Time	~minutes	hours-days
Lipid	Preserved	Dissolved (empty vacuoles)
Enzyme activity	Preserved	Lost
Morphology	Inferior, ice crystal artifacts	Superior, best detail

Feature	TEM	SEM	Light
Signal	Transmitted electrons, electron scatter	Secondary electrons from surface	Visible light absorption/transmission
Image	2-D internal	3-D surface	2-D color
Resolution	~0.1 nm	~1-10 nm	~0.2 μm

Best for	Organelles, filtration barrier, foot processes	Cilia, microvilli surface	Survey, H&E
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Feature	RER	SER
Ribosomes	Present	Absent
Function	Protein for export/membrane/lysosome	Lipid synthesis, detox, Ca ²⁺ storage
Stain	Basophilic	Not basophilic
Cell example	Plasma cell, pancreatic acinar	Hepatocyte, adrenal fasciculata, muscle

Final page - End of QA sample

Glyphs verified: μ μm α β ${}^3\text{H}$ $\pm$ $\rightarrow$ γ

If this text shows correctly, glyphs render without tofu.